PROJECT REPORT

Bike Sharing Data Analysis and Visualization Using Power BI

1. Student Details

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Project Title: Bike Sharing Dashboard

Software Used: Microsoft Power BI Desktop

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2. Dataset Description

The Bike Sharing dataset contains information about stations, bikes, users, and trip durations.

The dataset is used to analyze bike-sharing activities, identify popular stations, understand user behaviour, and discover patterns in bike usage.

Main dataset attributes include:

- Start Date and Start Station
- End Date and End Station
- Bike ID / Bike Number
- Trip Duration
- User Type

Dataset Overview:

Parameter	Details
Total Trips	Approximately 50,000
Unique Bikes	476
Starting Stations	50
Ending Stations	51
Average Trip Duration	17.75 minutes
Displayed Date Range	11-01-2014 to 04-12-2015

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3. Project Objectives

The main objectives of this project are:

1. To analyze bike-sharing usage and identify overall rental trends.
2. To identify the most popular starting and ending stations.
3. To analyze daily, monthly, and hourly bike rental patterns.
4. To compare bike usage between different user categories.
5. To calculate average trip duration and understand riding behaviour.
6. To identify the most frequently used bikes.
7. To develop an interactive dashboard using Microsoft Power BI for better data visualization and decision-making.

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4. Data Cleaning and Preprocessing

Data cleaning and preprocessing were performed to prepare the dataset for analysis and visualization.

The following transformations were used:

1. **Data Type Correction:** Ensured that date, time, numeric, and text columns had appropriate data types.
2. **Date Extraction:** Created a separate Trip_Date column from the original start date/time.
3. **Hour Extraction:** Created Trip_Hour to identify the hour of each bike trip.
4. **Day Extraction:** Created Day_Name to identify the day of the week.
5. **Month Extraction:** Created Month_Name and Month_Number for monthly analysis.
6. **Trip Duration Conversion:** Converted trip duration from seconds into minutes.
7. **Date Table Creation:** Created a separate DateTable using the CALENDAR function for time-based analysis.
8. **Data Modeling:** Established a relationship between the DateTable and the main trip dataset.

These transformations made the dataset suitable for analytical calculations and dashboard development.

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5. Data Model

The project uses a date-based data model to support time-based analysis.

Main Table: Trip

- Trip records
- Start and end dates
- Starting and ending stations
- Bike information

- User type
- Trip duration

Date Table: DateTable

- Date
- Year
- Month Name
- Month Number
- Day Name
- Day Number
- Quarter

A one-to-many relationship was established between DateTable[Date] and the date-only column in the Trip table.

This relationship enables date filtering and time-based analysis across the dashboard.

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6. DAX Measures and Calculated Columns

DAX (Data Analysis Expressions) was used to create calculated columns and measures for dashboard analysis.

A. Total Trips

Total Trips = COUNTROWS('trip')

B. Unique Bikes

Unique Bikes =

DISTINCTCOUNT('trip'[Bike number])

C. Average Trip Duration

Average Duration =

AVERAGE('trip'[Trip_Duration_Minutes])

D. Trip Duration in Minutes

Trip_Duration_Minutes =

'trip'[tripduration] / 60

E. Trip Hour

Trip_Hour =

HOUR('trip'[starttime])

F. Trip Date

```
Trip_Date =
DATE(
    YEAR('trip'[starttime]),
    MONTH('trip'[starttime]),
    DAY('trip'[starttime])
)
```

G. Month Name

```
Month_Name =
FORMAT('trip'[Trip_Date], "MMMM")
```

H. Day Name

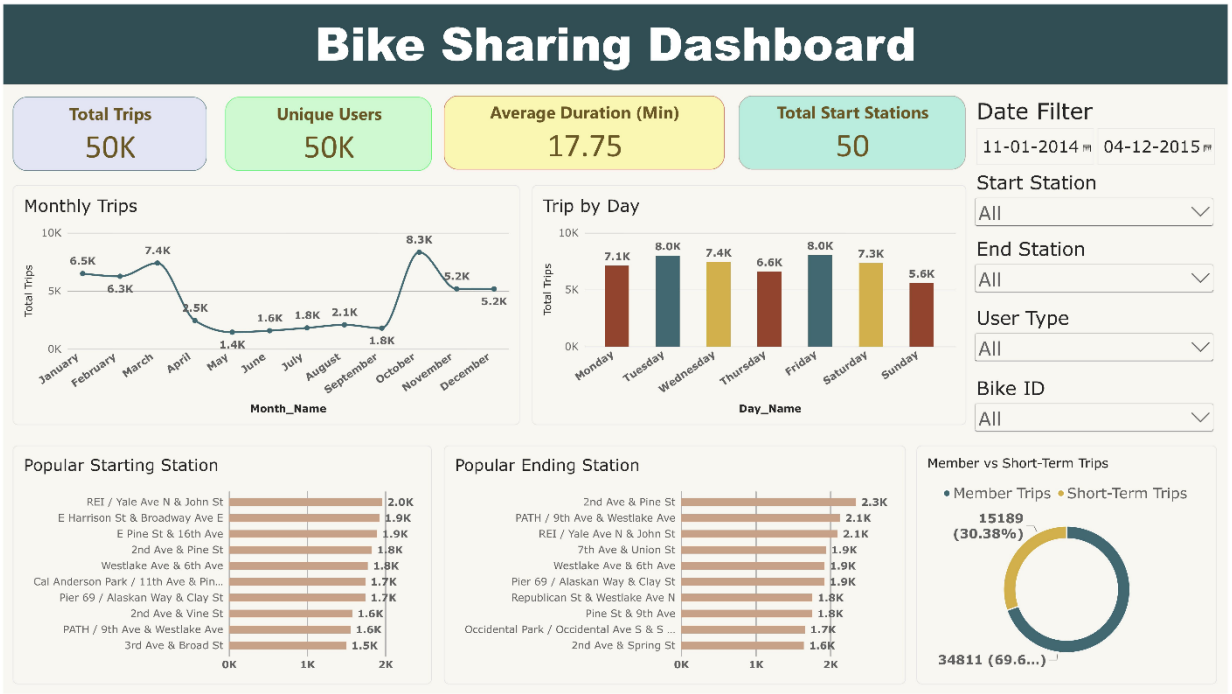
```
Day_Name =
FORMAT('trip'[Trip_Date], "dddd")
```

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7. Dashboard Screenshots and Visualizations

The Power BI dashboard consists of three main analytical pages and one summary section.

Page 1: Bike Sharing Overview

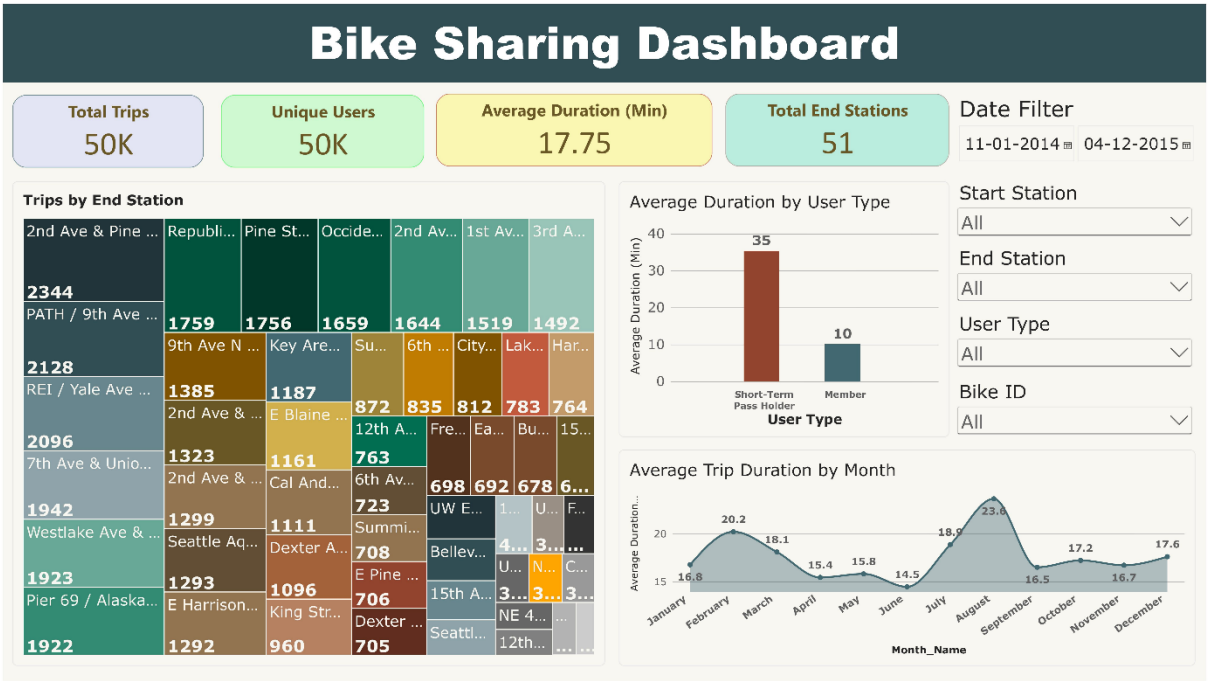


This page provides an overall view of bike-sharing activities.

Visualizations included:

- Total Trips Card
- Unique Users Card
- Average Duration Card
- Total Starting Stations Card
- Monthly Trips Trend
- Trips by Day
- Popular Starting Stations
- Popular Ending Stations
- Member vs Short-Term Trips
- Date and station filters

Page 2: Station and Trip Duration Analysis



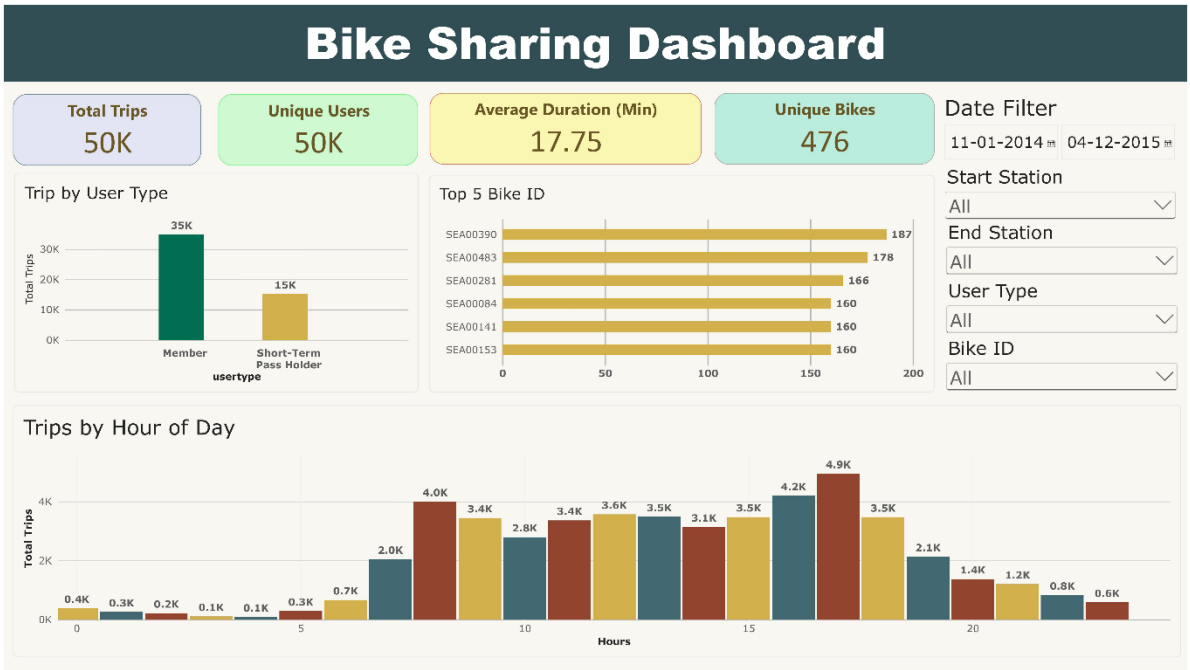
This page focuses on station performance and user riding behaviour.

Visualizations included:

- Trips by End Station
- Average Duration by User Type
- Average Trip Duration by Month
- Starting and ending station filters

- User Type and Bike ID filters

Page 3: User and Bike Analysis

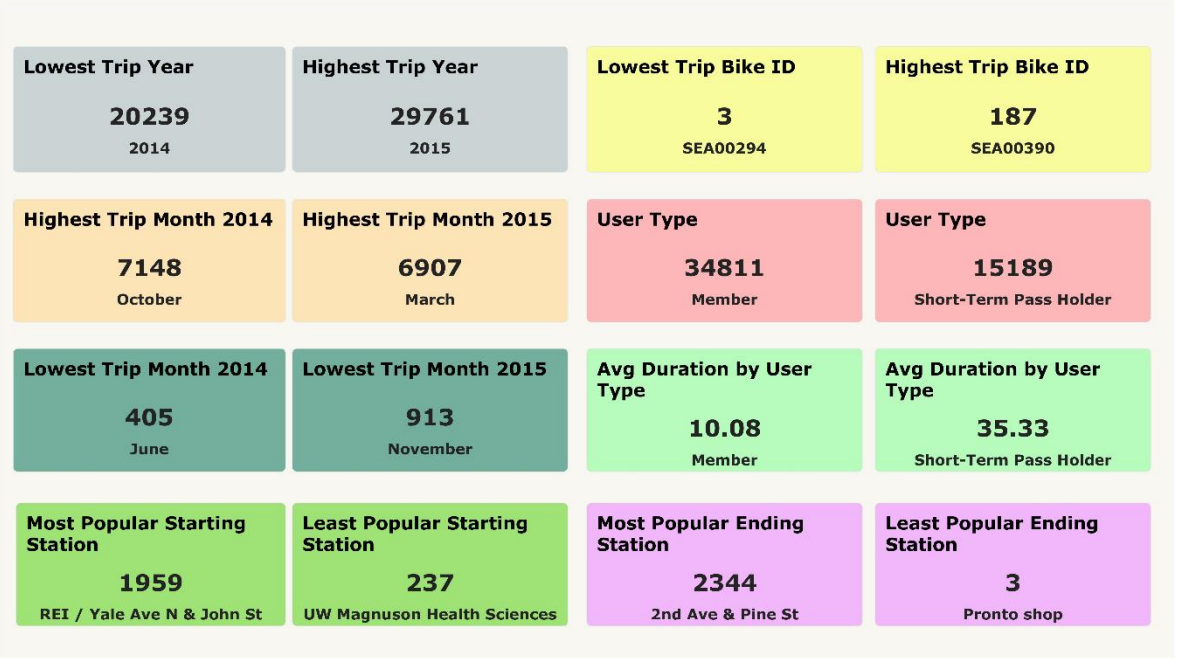


This page analyzes user categories and bike utilization.

Visualizations included:

- Trips by User Type
- Top 5 Bike IDs
- Trips by Hour of Day
- Unique Bikes KPI
- Date and category filters

Page 4: Key Performance Summary



This section displays important analytical findings, including:

- Highest and lowest trip years
- Highest and lowest trip months
- Highest and lowest performing bike IDs
- Average duration by user type
- Most and least popular starting stations
- Most and least popular ending stations

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8. Key Findings

Based on the dashboard analysis, the following findings were observed:

1. The dashboard records approximately 50,000 total bike trips.
2. A total of 476 unique bikes were identified.
3. The overall average trip duration is approximately 17.75 minutes.
4. Member users account for 34,811 trips (69.6%), while Short-Term Pass Holders account for 15,189 trips (30.38%).
5. October 2014 recorded the highest monthly trips in 2014, with 7,148 trips.
6. March 2015 recorded the highest monthly trips in 2015, with 6,907 trips.
7. June 2014 recorded the lowest monthly trips in 2014, with 405 trips, while November 2015 recorded the lowest in 2015, with 913 trips.
8. REI / Yale Ave N & John St was the most popular starting station, recording 1,959 trips.
9. 2nd Ave & Pine St was the most popular ending station, recording 2,344 trips.
10. SEA00390 was the most frequently used bike, with 187 trips.
11. The average trip duration for Members was 10.08 minutes, compared with 35.33 minutes for Short-Term Pass Holders.
12. The hourly analysis shows that bike usage is higher during daytime hours, with a peak around 8 PM (approximately 4,900 trips).

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9. Conclusion

The Bike Sharing Data Analysis project successfully demonstrates how Microsoft Power BI can be used to analyze and visualize bike-sharing data.

Through data cleaning, preprocessing, DAX calculations, data modeling, and interactive visualizations, the project provides useful insights into trip patterns, station popularity, bike utilization, and user behaviour.

The analysis reveals differences in usage between Member and Short-Term Pass Holder categories and identifies important trends across different months and hours.

Overall, this project demonstrates the practical application of data analytics and business intelligence tools in understanding real-world transportation data and supporting data-driven decision-making.

Future Scope:

The project can be further enhanced by integrating weather information, forecasting future bike demand, identifying station-level shortages, and developing predictive models for bike-sharing utilization.